Final Project:
Natural Disaster Severity Prediction

April 30, 2026

Data Mining

Spring 2026

Outline

- Introduction
- Dataset Overview
- Objective
- Rules
- Submission & Grading Criteria
- Reference

Introduction

The final project will be conducted as a Kaggle Competition.

- **It is an Group Work and counts for 35% of the semester grade**
- Final Project Kaggle Link

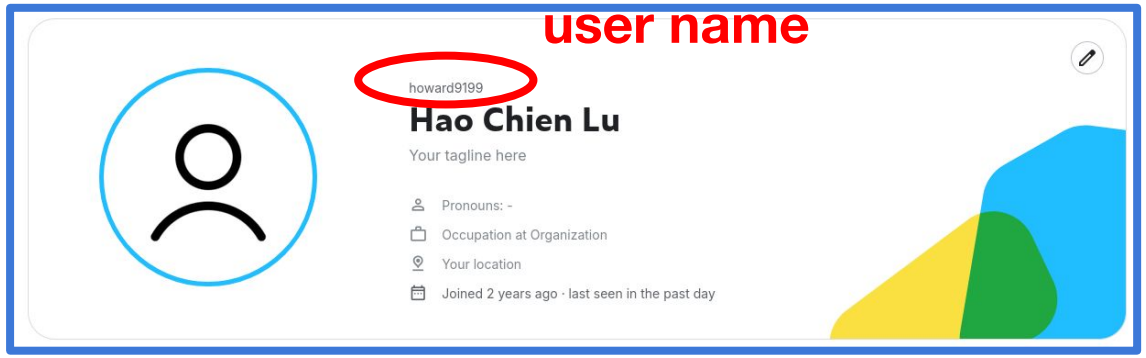
- Please sign in with your **NYCU email** account directly.

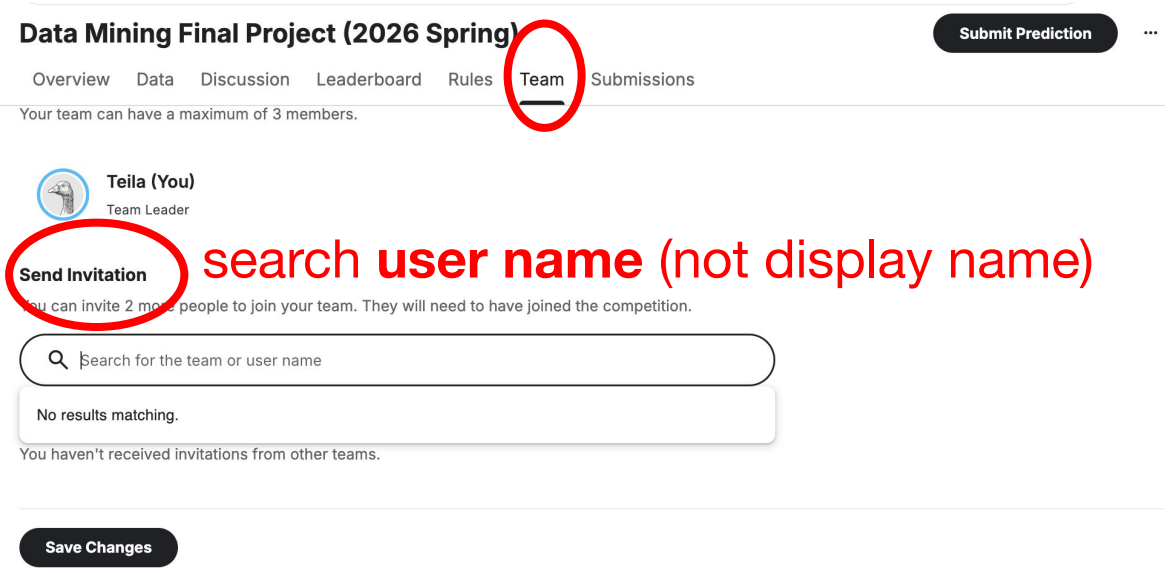
- **Deadline: June 10, 11:55 pm**
- **Core Task:**
  - Predict natural disaster severity levels for the next five weeks using only historical meteorological data.

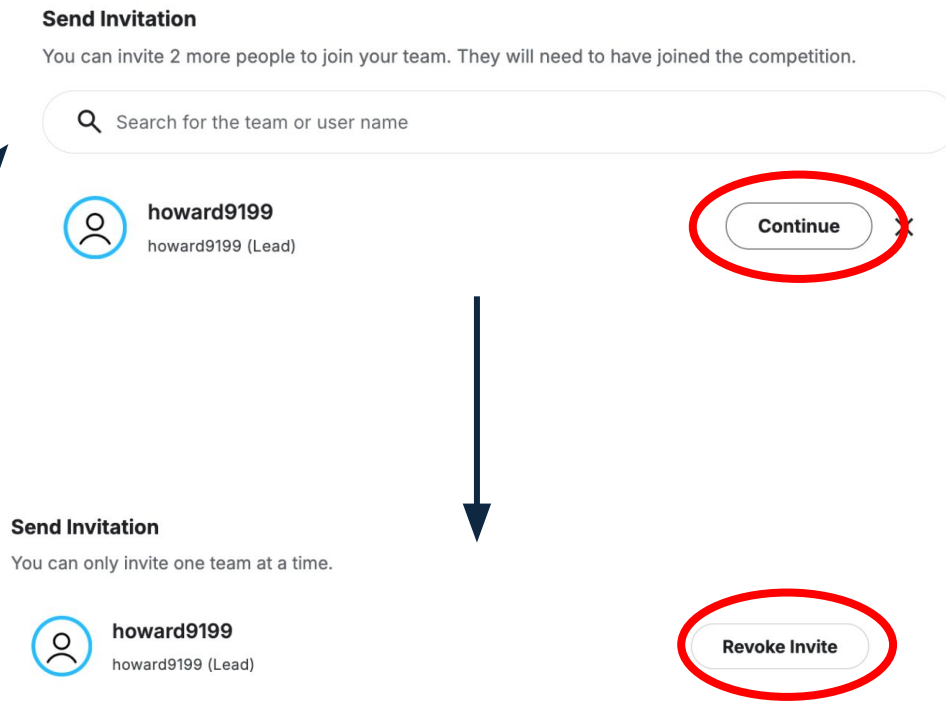
Introduction

Invite team members

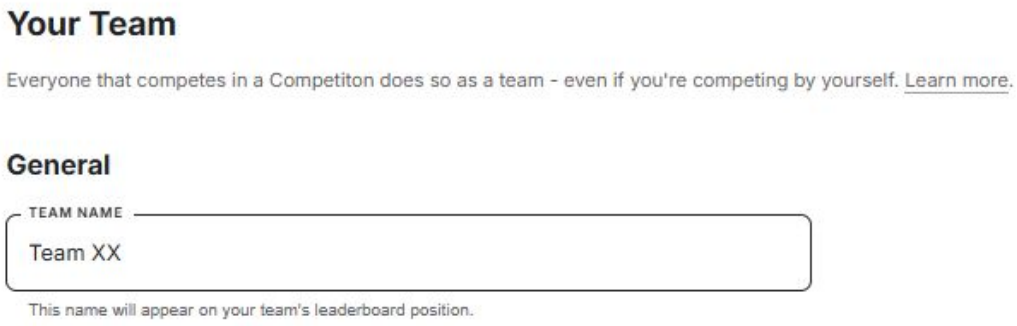
Team -> Send invitation -> Click continue







Kaggle Submission Rule



- **Name limitation**:

  Change the team name in Kaggle as **Team {Group ID}** (e.g. Team 1, Team 21).

- **Daily Limit**:

  Maximum of **3 submissions** per team per day (reset on UTC+0, i.e. 8 AM in Taiwan).

- **Evaluation**: Submissions are evaluated using **MAE (Mean Absolute Error)**.

- The private leaderboard will only be visible after the competition ends.
  - By default, your **two** best public leaderboard submissions will be used for the private leaderboard.

Dataset

We are using a Natural Disaster Severity Prediction Dataset:

- **Files**: Includes train.csv and test.csv.
- **Columns**:
  - `region_id`: Unique identifiers for different geographic regions.
  - `date`: Observation dates.
    - In "`train.csv`", each region has a total of 5,480 days.
  - Other Features: meteorological data

Indicator	Description
region_id	Region ID
date	Observation Date
wind	Wind Speed
wind_min	Minimum Wind Speed
wind_max	Maximum Wind Speed
wind_range	Wind Speed Range
humidity	Humidity
tmp	Temperature
tmp_range	Temperature Range
tmp_max	Maximum Temperature
tmp_min	Minimum Temperature
surf_tmp	Surface Temperature
surf_pre	Surface Pressure
dp_tmp	Dew Point Temperature
wb_tmp	Wet Bulb Temperature
prec	Precipitation

Dataset

- **Label** (the column "`score`"):
  - Ranges from 0 to 5, with higher values indicating greater severity of the disaster.
    - Natural disaster type: **Drought**
  - Scores are recorded <u>weekly</u>
    - The dataset contains NaN values for the other 6 days of the week.
    - The score values in the provided datasets are all integers. Your predictions can be either integers or floats.

7 days

7 days

	region_id	date	prec	surf_pre	humidity	tmp	dp_tmp	wb_tmp	tmp_max	tmp_min	tmp_range	surf_tmp	wind	wind_max	wind_min	wind_range	score
0	R1	3004-12-31	0.00	101.27	3.76	5.89	-0.14	-0.11	14.87	-2.37	17.24	4.23	1.45	2.58	0.24	2.34	NaN
1	R1	3005-01-01	0.00	101.26	5.37	8.81	4.59	4.60	17.28	1.14	16.14	7.96	1.83	2.38	1.18	1.21	NaN
2	R1	3005-01-02	0.01	100.81	9.32	13.09	12.74	12.74	17.38	8.21	9.17	12.96	1.96	2.49	1.39	1.09	NaN
3	R1	3005-01-03	0.02	100.51	11.40	16.01	16.08	16.08	19.24	13.77	5.47	15.82	2.13	2.67	1.74	0.93	NaN
4	R1	3005-01-04	1.93	100.16	12.20	17.98	17.04	17.04	22.85	14.27	8.58	17.86	2.91	4.25	2.13	2.11	NaN
5	R1	3005-01-05	15.67	100.34	8.63	12.24	10.74	10.75	17.85	3.09	14.76	12.10	2.96	4.06	2.02	2.03	NaN
6	R1	3005-01-06	0.01	101.59	2.80	1.32	-3.97	-3.90	6.66	-4.29	10.95	1.41	3.15	4.17	2.00	2.17	0.0
7	R1	3005-01-07	0.00	101.86	1.85	-2.33	-9.02	-8.75	4.80	-7.23	12.03	-2.57	2.00	2.38	1.55	0.83	NaN
8	R1	3005-01-08	5.37	101.17	3.20	0.35	-2.49	-2.44	2.53	-3.66	6.19	0.45	1.36	1.92	0.60	1.33	NaN
9	R1	3005-01-09	3.99	100.59	4.42	3.24	2.06	2.07	6.45	-0.76	7.21	3.23	2.14	3.61	0.17	3.45	NaN

Objective

- Task:

  Train a model based on "`train.csv`", then use the 91 days of data for each region provided in "`test.csv`" as input to <u>predict each region's disaster severity levels</u> for <u>**the next five weeks**</u>.

Training Data Format

Train.csv and region_id = R1

	Region ID	Date	XXX	Score
	R1	XXXX-09-18	XXX	NAN
91 days	. . . .	. . . .	. . . .	. 2 . .
	R1	XXXX-12-17	XXX	1
	R1	XXXX-12-18	XXX	NAN
35 days	. . . . R1	. . . .	. . . .	1 . 2 . 1 . 2 . 1

Model

Predict the score columns for **next 5 consecutive weeks (after 91 days)**

Testing Data Format

Test.csv

no "`score`" column

91 days

91 days

Region ID	Date	XXX
R1	XXXX-09-18	XXX
. . . .	. . . .	. . . .
R1	XXXX-12-17	XXX
R2	XXXX	XXX
. . .	. . .	. . .

Model

Predict the score columns for **next 5 consecutive weeks (after 91 days)**

Kaggle Submission Format

- **Format:** We provide a template submission file (“`sample_submission.csv`”).

Inference

R1_data in test.csv → Model → R1_future_score [1.5, 2.0, 2.1, 0.2, 1.2]

submission.csv

region_id,	pred_week1,	pred_week2,	pred_week3,	pred_week4,	pred_week5
R1 ,	1.5,	2.0,	2.1,	0.2,	1.2
R2 ,	0.5,	0.8,	1.2,	1.5,	0.7
R3 ,	3.0,	3.1,	3.5,	0.6,	0.9

Grading Criteria

The grade for the **Final Project [35 points]** is based on:

1. **Kaggle & Report [30 points]**
   a. Kaggle Competition (60%)
   b. Your Report (40%)
2. **Progress Check** (May 21) **[5 points]**

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Grading Criteria - Kaggle & Report **[30 points]**

- **Kaggle Competition (60%)**
  - 30% based on Public Leaderboard Points (40% of testing data)
  - 70% based on Private Leaderboard Points (remaining 60% of testing data)

- **Three Baselines**

Public Leaderboard
(Private Leaderboard is hidden until the end of the competition)

#	Team	Members	Score
	Baseline 3		0.8056
	Baseline 2		0.8623
	Baseline 1		0.9117

Grading Criteria - Kaggle & Report **[30 points]**

- **Kaggle Competition (60%)**
  - 30% based on Public Leaderboard Points
  - 70% based on Private Leaderboard Points

Leaderboard Point Calculation

Your Model Performance	Points	Calculation Method
Above Baseline 3	42 ~ 60	$42 + 18 \times \frac{N - Rank}{N - 1}$ **N: the number of people over Baseline 3 (If N=1, then the point = 60)**
Between Baselines 2 & 3	25 ~ 40	$25 + 15 \times \frac{YOUR_Score - B2_Score}{B3_Score - B2_Score}$
Between Baselines 1 & 2	10 ~ 25	$10 + 15 \times \frac{YOUR_Score - B1_Score}{B2_Score - B1_Score}$
Under Baseline 1	5	5

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Grading Criteria - Kaggle & Report **[30 points]**

- **Report (40%)**
  - **Format**
    - Please use the provided Overleaf template to write your report
      - Overleaf template is built on IEEE paper template
      - You may also use the Microsoft Word template provided by IEEE (please download the **A4 format** version)

    - Modification of the basic format is not allowed (e.g., font size, line spacing).

Grading Criteria - Kaggle & Report [30 points]

- **Report (40%)**
  - **Content (Total 5-8 Pages, Not Include References)**
    - i. **Abstract:** Include your group ID and Github link (with README.md in Github repo)
      - README.md: Describe in detail how to run your code.
    - ii. **Project Summary (Max 1 Page)**
      - Describe the goal of this project
      - Provide the high-level idea of your method
      - Introduce the related works
    - iii. **Proposed Method (1-3 Pages)**
      - Please describe your proposed method in detail. You may include formulas, pseudocode, framework diagrams, etc.
    - iv. **Experiments (2-4 Pages)**
      - Data description, statistics of data, and the basic setup for your experiments.
      - You can also include your self-defined baselines to reveal the robustness of your method.
      - We encourage you to include experiments and analysis from different perspectives, such as
        - Your model's performance on Kaggle (you can compare current and past results)
        - Comparison with your self-defined baselines
        - Ablation study, Parameter analysis, Case study, etc.

Grading Criteria

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1. **Kaggle & Report [30 points]**
   a. Kaggle Competition (60%)
   b. Your Report (40%)
2. **Progress Check** (May 21) **[5 points]**

Grading Criteria - Progress Check **[5 points]**

- **Date:** May 21
- **Progress Check:**
  - **<u>5-minute</u>** presentation for each group
  - You may briefly describe your current observations and any challenges you have encountered.
  - Grading will be based on the completeness of the progress you present.

  - There is no need to send the slides to the TAs in advance, but please **<u>remember to bring your own laptop</u>** for the presentation.

Grading Criteria

- You are allowed to use the AI tools to assist your work, but …
  - Any form of plagiarism in code or report will result in a **score of 0 for this assignment.**
  - The submitted **report, code, and Kaggle results must be consistent and reproducible**.
    - If the code cannot be executed, or if the Kaggle results are inconsistent with the submitted code or report, this will also result in a **score of 0 for this assignment**.
  - If any **hallucinations** are found in the report (e.g., non-existent references), **the report will receive a score of 0.**

If there are concerns about your submission,
TAs may require an on-site demonstration to verify the results.

Report and Code Submission

Please submit your report & code by **June 10, 11:55 PM.**

Late submissions are not accepted.

Submissions consist of two parts (Besides Kaggle Submission):

1. **Code**
   a. Please host your **code on a public GitHub repository** (commit before deadline).
2. **Report.**
   a. Please ensure your public **GitHub link is included** within your report, and submit your **report via E3.**
   b. **File name:** DM_project_Group_{GroupID}.pdf (e.g., DM_project_Group_17.pdf)

Please write down your report in English.

The TA will test whether your code can run successfully; additionally, your **Kaggle Display Name** must be **Team {Group ID}**, or your grade will be recorded as 0.

If you have any other questions, please feel free to contact the TA at nycu.dm.ta@gmail.com.

Optional Oral Presentation

- **Date:** June 11 (Final week of the DM course)

- **Grading Criteria** [Up to 5% extra credit to the semester grade]:
  - Based on the completeness of your final project and your presentation

- **First come, First served:**
  - Due to limited class time, groups that sign up earlier will have priority to present.

- Sign up link and other information will be released later.

Reference

- **Here are some papers related to other Kaggle competitions for your reference**

  - Iglovikov, Vladimir, Sergey Mushinskiy, and Vladimir Osin. "Satellite imagery feature detection using deep convolutional neural network: A kaggle competition." *arXiv preprint arXiv:1706.06169* (2017).

  - Kechyn, Glib, et al. "Sales forecasting using WaveNet within the framework of the Kaggle competition." *arXiv preprint arXiv:1803.04037* (2018).